

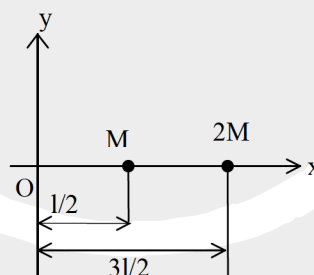
**HINTS AND SOLUTIONS**

- Q.1 A  
Q.2 A  
Q.3 B  
Q.4 C  
Q.5 B  
Q.6 C  
Q.7 D  
Q.8 B  
Q.9 C  
Q.10 B  
Q.11 D  
Q.12 A  
Q.13 A  
Q.14 A  
Q.15 B  
Q.16 D  
Q.17 C  
Q.18 A  
Q.19 B  
Q.20 C  
Q.21 C  
Q.22 D

Sol. Considering the c.m. of the uniform sticks at their geometrical centres we find the c.m. of the composite stick as

$$X_{cm} = \frac{M\left(\frac{l}{2}\right) + (2M)\left(\frac{3l}{2}\right)}{M + 2M}$$

$$\Rightarrow X_{cm} = \frac{3.5Ml}{3M} \cong 1.16 \ell$$



Q.23  $\left[ b - \frac{l}{\ln\left(\frac{b}{b-l}\right)} \right]$

Q.24 Ans :  $\sqrt{3} r$

Q.25 Ans : 115.55 m